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Method and Apparatus for Adjusting Natural Language Sentence

Abstract

Embodiments of this application disclose a method and an apparatus for adjusting a natural language sentence, and a storage medium. The method includes: receiving a natural language sentence including a placeholder, where the placeholder includes a type attribute, and the placeholder is occupied by a sentence constituent matching the type attribute; determining a position of the placeholder in the natural language sentence; and adding a preposition to the natural language sentence based on the type attribute of the placeholder and the position of the placeholder in the natural language sentence. The preposition is automatically added to the natural language sentence through the placeholder, so that readability and accuracy of the natural language sentence can be improved. In addition, when a word order of the natural language sentence is adjusted, the preposition is automatically updated correspondingly, which ensures the readability and the accuracy of the sentence.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a U.S. National Stage Application of International Application No. PCT/CN2022/090003 filed Apr. 28, 2022, which designates the United States of America, the contents of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] Teachings of the present disclosure relate to the field of natural language sentence processing technologies. Various embodiment of the teachings herein include methods and/or apparatus for adjusting a natural language sentence.

BACKGROUND

[0003] Natural language generation (NLG) is a branch of artificial intelligence and computational linguistics. A corresponding language generation system is a computer model based on language information processing, which operates in a process opposite to natural language analysis, starts from an abstract concept hierarchy, and generates a text by selecting and executing specific semantic and grammatical rules. The NLG technology automatically generates an understandable natural language text by using artificial intelligence and linguistics method. The NLG technology reduces difficulty in communication between a human and a computer, and is widely applied to fields such as machine news writing and chatting robotics, and has become one of research hotspots of artificial intelligence. There are opportunities to improve readability and accuracy of a generated natural language sentence.

SUMMARY

[0004] Embodiments of this application provide method and apparatus for adjusting a natural language sentence. For example some embodiments include a method for adjusting a natural language sentence comprising: receiving a natural language sentence including a placeholder, where the placeholder includes a type attribute, and the placeholder is occupied by a sentence constituent matching the type attribute; determining a position of the placeholder in the natural language sentence; and adding a preposition to the natural language sentence based on the type attribute of the placeholder and the position of the placeholder in the natural language sentence. Through the addition of the preposition in the natural language sentence through the type attribute of the placeholder and the position of the placeholder in the natural language sentence, readability and accuracy of the natural language sentence can be improved.

[0005] In some embodiments, adding a preposition to the natural language sentence based on a type of the placeholder and the position of the placeholder in the natural language sentence includes: determining a first sentence constituent occupying a subject placeholder in the natural language sentence as a subject of the natural language sentence; determining a second sentence constituent occupying a predicate placeholder in the natural language sentence as a predicate of the natural language sentence; determining a third sentence constituent occupying an object placeholder in the natural language sentence as an object of the natural language sentence; adding a first preposition between the subject and the object when the object is located between the subject and the predicate; and adding a second preposition between the object and the subject when the object is located before the subject and the predicate. It may be learned that different types of prepositions may be added based on a position of the object, which is more in line with a natural language habit.

[0006] In some embodiments, adding a preposition to the natural language sentence based on a type of the placeholder and the position of the placeholder in the natural language sentence includes:

determining a fourth sentence constituent occupying a complement placeholder in the natural language sentence as a first complement of the natural language sentence, where the first complement is determined based on a user input; and adding a third preposition before the first complement. It may be learned that the preposition may be added for the complement determined based on the user input.

[0007] In some embodiments, adding a preposition to the natural language sentence based on a type of the placeholder and the position of the placeholder in the natural language sentence includes: determining a fifth sentence constituent occupying the complement placeholder in the natural language sentence as a second complement of the natural language sentence, where the second complement is determined based on output data of a service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and adding a fourth preposition before the second complement. It may be learned that the preposition may be added for the complement determined based on the output data of the service operation.

[0008] In some embodiments, adding a preposition to the natural language sentence based on a type of the placeholder and the position of the placeholder in the natural language sentence includes: determining a sixth sentence constituent occupying an adverbial placeholder in the natural language sentence as an adverbial of the natural language sentence, where the adverbial is determined based on input data of the service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and adding a fifth preposition before the adverbial. The prepositions may be added for a plurality of types of adverbials, which is more in line with the natural language habit.

[0009] In some embodiments, the method further includes: moving the placeholder in the natural language sentence based on a user-triggered instruction; determining an updated position of the placeholder in the natural language sentence; and updating the preposition in the natural language sentence based on the type attribute of the placeholder and the updated position of the placeholder in the natural language sentence. When the placeholder of the natural language sentence is moved, the preposition is updated correspondingly, thereby ensuring the readability and the accuracy of the sentence.

[0010] In some embodiments, the natural language sentence is adapted to describe a workflow. The workflow represents that a resource in a workcell performs the service operation. Therefore, the preposition may be added to the natural language sentence describing a workflow, thereby improving execution efficiency of the workflow.

[0011] In some embodiments, the method further includes: converting the natural language sentence into a low-code representation describing the workflow; parsing the low-code representation, to obtain a workflow represented in a form of node link assembly; and compiling and downloading the workflow represented in the form of the node link assembly to runtime of a main controller of a corresponding workcell, to execute the workflow. The natural language sentence describing the workflow is converted into the low-code representation executing the workflow, which may improve processing efficiency of the workflow.

[0012] As another example, some embodiments include an apparatus for adjusting a natural language sentence comprising: a receiving module, configured to receive a natural language sentence including a placeholder, where the placeholder includes a type attribute, and the placeholder is occupied by a sentence constituent matching the type attribute; a determining module, configured to determine a position of the placeholder in the natural language sentence; and an addition module, configured to add a preposition to the natural language sentence based on the type attribute of the placeholder and the position of the placeholder in the natural language sentence.

[0013] In some embodiments, the addition module is configured to: determine a first sentence constituent occupying a subject placeholder in the natural language sentence as a subject of the

natural language sentence; determine a second sentence constituent occupying a predicate placeholder in the natural language sentence as a predicate of the natural language sentence; determine a third sentence constituent occupying an object placeholder in the natural language sentence as an object of the natural language sentence; add a first preposition between the subject and the object when the object is located between the subject and the predicate; and add a second preposition between the object and the subject when the object is located before the subject and the predicate.

[0014] In some embodiments, the addition module is configured to: determine a fourth sentence constituent occupying a complement placeholder in the natural language sentence as a first complement of the natural language sentence, where the first complement is determined based on a user input; and add a third preposition before the first complement.

[0015] In some embodiments, the addition module is configured to: determine a fifth sentence constituent occupying the complement placeholder in the natural language sentence as a second complement of the natural language sentence, where the second complement is determined based on output data of a service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and add a fourth preposition before the second complement.

[0016] In some embodiments, the addition module is configured to: determine a sixth sentence constituent occupying an adverbial placeholder in the natural language sentence as an adverbial of the natural language sentence, where the adverbial is determined based on input data of the service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and add a fifth preposition before the adverbial.

[0017] In some embodiments, the apparatus further includes an update module configured to: move the placeholder in the natural language sentence based on a user-triggered instruction; determine an updated position of the placeholder in the natural language sentence; and update the preposition in the natural language sentence based on the type attribute of the placeholder and the updated position of the placeholder in the natural language sentence.

[0018] In some embodiments, the natural language sentence is adapted to describe a workflow. The workflow represents that a resource in a workcell performs the service operation.

[0019] In some embodiments, the apparatus further includes an execution module, configured to: convert the natural language sentence into a low-code representation describing the workflow; parse the low-code representation, to obtain a workflow represented in a form of node link assembly; and compile and download the workflow represented in the form of the node link assembly to runtime of a main controller of a corresponding workcell, to execute the workflow.

[0020] As another example, some embodiments include an apparatus for adjusting a natural language sentence comprising: at least one memory, configured to store computer-readable code; and at least one processor, configured to invoke the computer-readable code to perform the steps of the above method for adjusting a natural language sentence in any of the above implementations.

[0021] As another example, some embodiments include a computer-readable storage medium storing computer-readable instructions. The computer-readable instructions, when executed by a processor, cause the processor to perform one or more of the methods described herein for adjusting a natural language sentence.

[0022] In some embodiments, a computer program product is stored in a tangible computer-readable medium, and includes computer-readable instructions. The computer-readable instructions, when executed, cause at least one processor to perform one or more of the methods for adjusting a natural language sentence.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 is a flowchart of an example method for adjusting a natural language sentence incorporating teachings of the present disclosure;

[0024] FIG. 2A shows a schematic diagram for describing an example workflow in a low-code architecture in a no-code manner incorporating teachings of the present disclosure;

[0025] FIG. 2B is a schematic diagram of an interaction based on a language element library incorporating teachings of the present disclosure;

[0026] FIG. 2C is a schematic diagram of a universal language word order template of a standard operating procedure (SOP) incorporating teachings of the present disclosure;

[0027] FIG. 2D is a schematic diagram of an architecture for automatically adjusting a word order position incorporating teachings of the present disclosure;

[0028] FIG. 2E is a schematic diagram of an example for automatically adjusting a word order incorporating teachings of the present disclosure;

[0029] FIG. 2F is a schematic diagram of an example overall semantic editing process incorporating teachings of the present disclosure;

[0030] FIG. 3 is an application scenario of a method for adjusting a natural language sentence in an industrial automation field incorporating teachings of the present disclosure;

[0031] FIG. 4 is a structural diagram of an example apparatus for adjusting a natural language sentence incorporating teachings of the present disclosure; and

[0032] FIG. 5 is a structural diagram of an example apparatus for adjusting a natural language sentence incorporating teachings of the present disclosure.

[0033] Reference numerals are as follows:

TABLE-US-00001 100 Method for adjusting a natural language sentence 101 to 103 Step 302 No-code universal language editor 303 No-code semantic standard operating procedure (SOP) editor 304 Universal runtime engineering exchange model 305 Low-code behavior tree/function block type diagram (FBTD) 306 Interpreter-driven runtime 307 Logic library 308 Resource library 309 Function block library 310 Customized thesaurus 311 Bill of materials (BOM) 312 Universal language extension thesaurus 313 Keypoint library 400 Language element library 401 Subject placeholder 402 Predicate placeholder 403 Adverbial placeholder 404 Object placeholder 405 Logic placeholder 406 Annotation placeholder 407 Semantic expression placeholder 408 Exception placeholder 409 Keypoint placeholder 410 Figure placeholder 411 Video placeholder 412 Attributive placeholder 413 Complement placeholder 414 Automatic generation processing 415 Automatic label generation processing 416 Automatic preposition generation processing 417 Label 418 Subject thesaurus 419 Natural language sentence after a subject is determined 420 Natural language sentence after a predicate placeholder is dragged 421 Predicate thesaurus 422 Natural language sentence after a predicate is determined 431 SOP template 432 Rule-based linking engine 433 Preposition 434 Prepositive preposition 435 Postpositive preposition 436 Specific language grammar 437 Chinese semantic thesaurus 438 English semantic thesaurus 450 Complement-to-subject conversion processing 750 Subject placeholder dragging processing 751 Subject determining and predicate placeholder dragging processing 752 Predicate determining processing 753 Object placeholder dragging and first preposition generation processing 754 Object determining processing 755 Adverbial placeholder dragging, adverbial determining, and second preposition generation processing 756 Complement placeholder dragging, complement determining, and second preposition generation processing 757 Attributive placeholder dragging and attributive determining processing 758 Word order adjustment processing 10 low-code development tool of an operational technology (OT) domain 20 Microservice generator of the OT domain 30 Runtime on a main controller of a workcell 200 Knowledge platform 301 Code development tool of an intelligence technology (IT) domain 700 Apparatus for adjusting a natural language sentence 701 Receiving module 702 Determining module 703 Addition module 704

DETAILED DESCRIPTION

[0034] Discussion of the various embodiments of the teachings herein is intended to make a person skilled in the art better understand and implement the subject described herein, and is not intended to limit the protection scope of the disclosure, the applicability, or examples. Changes may be made to functions and arrangements of the discussed elements without departing from the protection scope of the content of the embodiments of this disclosure. Various processes or constituents may be omitted, replaced, or added in each example as required. For example, the described method may be performed based on an order different from the order described herein, and steps may be added, omitted, or combined. In addition, features described with respect to some examples may also be combined in another example.

[0035] As used herein, the term “include” and its variants represent open terms, and mean “including but not limited to”. The term “based on” means “at least partially based on”. Terms “one embodiment” and “an embodiment” mean “at least one embodiment”. A term “another embodiment” means “at least one other embodiment”. Terms “first”, “second”, and the like may represent different objects or the same object. Other definitions may be included explicitly or implicitly in the following. Unless otherwise clearly specified in the context, a definition of a term is consistent throughout the specification.

[0036] Currently, a processing manner in which a natural language sentence is generated by using artificial intelligence such as machine learning or deep learning exists. The applicant finds that, the artificial intelligence generation manner requires a large number of computing resources, has high operating complexity, and is often not suitable for an application scenario that is sensitive to computing resources and operating complexity (for example, an industrial site). In addition, ensuring readability and accuracy of the natural language sentence without significantly occupying computing resources is also a problem.

[0037] Teachings of the present disclosure include methods and apparatus for adjusting a natural language sentence. Through automatically addition of a preposition to a natural language sentence determined based on a placeholder filling operation readability and accuracy of the natural language sentence may be improved.

[0038] The “preposition” herein is a vocabulary or an affix used for expressing a grammatical function of a word in grammar. The preposition is generally used before a pronoun or a noun phrase, and forms a preposition structure with the words, to represent a word such as a place, time, a state, a manner, a reason, a purpose, or a comparison object.

[0039] For example, the preposition may include: (1). a preposition positioned before a time or a place, such as since, in, to, toward, at, when, toward, with, along, following, or the like; (2). a preposition positioned before a manner, such as according to, on the basis of, in accordance with, in line with, by way of, through, based on, by, by means of, or the like; (3). a preposition positioned before a purpose, such as for, for the sake of, or the like; (4). a preposition positioned before a reason, such as due to, since, because, only, or the like; (5). a preposition positioned before an object or content, such as to, for, by, from, following, with, about, or the like; (6). a preposition positioned before excluding, such as except, besides, unless, or the like; (7). a preposition positioned before passivity, such as is/are, named, allowed, given, enabled, or the like; (8). a preposition positioned before comparison, such as than, to, as, or the like; and (9). a preposition positioned before an identity: such as as and the like.

[0040] FIG. 1 is a flowchart of an example method for adjusting a natural language sentence incorporating teachings of the present disclosure. As shown in FIG. 1, the method **100** includes:

[0041] Step **101**: Receive or obtain a natural language sentence including a placeholder, where the placeholder includes a type attribute, and the placeholder is occupied by a sentence constituent matching the type attribute.

[0042] Step **102**: Determine a position of the placeholder in the natural language sentence.

[0043] Step **103**: Add a preposition to the natural language sentence based on the type attribute of the placeholder and the position of the placeholder in the natural language sentence.

[0044] In some embodiments, receiving or obtaining a natural language sentence including a placeholder in step **101** may be receiving or obtaining, for example, a natural language sentence, a paragraph of the natural language sentence, or a discourse-level natural language sentence inputted through a natural language compilation tool. In some embodiments, the natural language compilation tool may provide a human-computer interaction interface including a plurality of placeholders. The natural language compilation tool receives a user dragging operation through the human-computer interaction interface, to select the placeholder corresponding to the sentence constituent, and move the position of the placeholder in the natural language sentence. Moreover, the natural language compilation tool may select from a thesaurus through the human-computer interaction interface or directly provide user input, to determine the corresponding sentence constituent occupying placeholder.

[0045] In some embodiments, the receiving a natural language sentence including a placeholder in step **101** further includes: receiving, from the human-computer interaction interface, the natural language sentence that is generated based on a deep learning algorithm and includes the placeholder.

[0046] A specific implementation of receiving the natural language sentence is exemplarily described above. A person skilled in the art may realize that the description is merely exemplary and is not intended to limit the protection scope of the implementations of the teachings of the present disclosure.

[0047] In some embodiments, step **103** includes: determining first sentence constituent occupying a subject placeholder in the natural language sentence as a subject of the natural language sentence; determining a second sentence constituent occupying a predicate placeholder in the natural language sentence as a predicate of the natural language sentence; determining a third sentence constituent occupying an object placeholder in the natural language sentence as an object of the natural language sentence; adding a first preposition (for example, by) corresponding to an active voice between the object and the subject when the object is located between the subject and the predicate; and adding a second (for example, preposition is/are) corresponding to a passive voice between the object and the subject when the object is located before the subject and the predicate.

[0048] In some embodiments, step **103** includes: determining a fourth sentence constituent occupying a complement placeholder in the natural language sentence as a first complement of the natural language sentence, where the first complement is determined based on a user input; and adding a third preposition (for example, from or to) before the first complement.

[0049] In some embodiments, step includes: determining a fifth sentence constituent occupying the complement placeholder in the natural language sentence as a second complement of the natural language sentence, where the second complement is determined based on output data of a service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and adding a fourth preposition (for example, from or to) before the second complement.

[0050] When the first complement is a starting object of a service operation, the third preposition is a preposition (for example, from) adapted to the starting object. When the first complement is an end object of the service operation, the third preposition is a preposition (for example, to) adapted to the end object.

[0051] In some embodiments, step **103** includes: determining a sixth sentence constituent occupying an adverbial placeholder in the natural language sentence as an adverbial of the natural language sentence, where the adverbial is determined based on input data of the service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and add a fifth preposition (for example, to) before the adverbial.

[0052] In some embodiments, a change instruction (for example, a touch instruction for an automatically generated preposition on an interface) for changing an automatically generated preposition is received, a preposition thesaurus is displayed in response to the change instruction, so that a user selects a change preposition to replace the automatically generated preposition, and replace the automatically generated preposition with the change preposition. It may be learned that the automatically generated preposition may be modified based on the user instruction, thereby further improving readability.

[0053] In some embodiments, the method **100** further includes: moving the placeholder in the natural language sentence based on a user-triggered instruction; determining an updated position of the placeholder in the natural language sentence; and updating the preposition in the natural language sentence based on the type attribute of the placeholder and the update position of the placeholder in the natural language sentence.

[0054] In some embodiments, the method **100** further includes: the natural language sentence is adapted to describe a workflow. The workflow represents that a resource in a workcell performs the service operation. In some embodiments, the method **100** for adjusting a natural language sentence further includes: converting the natural language sentence into a low-code representation describing the workflow; parsing the low-code representation, to obtain a workflow represented in a form of node link assembly; and compiling and downloading the workflow represented in the form of the node link assembly to runtime of a main controller of a corresponding workcell, to execute the workflow.

[0055] Some embodiments include a method for describing a workflow (for example, an information technology (IT)/operational technology (OT) process) as a natural language sentence by using semantic language elements. The semantic language elements may come from a predetermined thesaurus. Through the method, a semantic standard operating procedure (SOP) or a more commonly used local language can be inputted into a low-code platform, and a corresponding low-code workflow can be executed finally at runtime. Correspondingly, definitions of the semantic language elements are extended, to provide support for necessary low-code constituents such as a behavior tree logic, a semantic expression, an exception, a keypoint, an annotation, a figure, a video, and an index marking.

[0056] Therefore, bidirectional semantic conversion between a no-code representation and a low-code representation (for example, a function block type diagram (FBTD) and an Internet of Things (IoT) behavior tree) becomes possible.

[0057] Some embodiments include a method for generating a natural language sentence describing a workflow. The method for adjusting a natural language sentence shown in FIG. 1 may be applied to the natural language sentence provided and describing the workflow for adjustment.

[0058] In some embodiments, the method for generating a natural language sentence describing a workflow includes:

[0059] Step 1: Determine a subject placeholder based on a first selection operation performed on a language element library. Herein, the language element library includes a plurality of types of placeholders. For example, the plurality of types of placeholders are listed in the language element library on a human-computer interaction interface in forms of icons. Each placeholder includes a type attribute, and each placeholder is adapted to be occupied by a sentence constituent matching the type attribute.

[0060] For example, sentence constituents of modern Chinese usually include a subject, a predicate, an object, a verb, an attributive, an adverbial, a complement, a head, and the like. Therefore, a language element library of Chinese may include a subject placeholder, a predicate placeholder, an object placeholder, a verb placeholder, an attributive placeholder, an adverbial placeholder, a complement placeholder, a head placeholder, and the like. Similarly, sentence constituents of an English sentence usually include a subject, a predicate, an object, a predicative, an attributive, an adverbial, a complement, and an appositive. Correspondingly, a language element

library of English may include a subject placeholder, a predicate placeholder, an object placeholder, a predicative placeholder, an attributive placeholder, an adverbial placeholder, a complement placeholder, and an appositive placeholder. For another language or local language, similar placeholders may be set based on a language habit and a grammar.

[0061] In some embodiments, a user may first select the subject placeholder from the language element library through, for example, a human-computer interaction interface, and drag the subject placeholder to a natural language sentence editing interface to determine the subject placeholder.

[0062] Step 2: Determine, from a subject thesaurus associated with the subject placeholder, a subject occupying the subject placeholder, where the subject corresponds to a resource. For example, the resource may be a resource in a workcell or may be a resource in a resource library in another type of low-code platform. The workcell may be a combination of resources such as a system or a device that can achieve a relatively complete and independent control process and operation. Workflow creation is performed by using the workcell as a basic cell, which is more in line with the characteristics of industrial control, can improve development integration, and reduce development complexity. For example, in the field of industrial technologies, the workcell may be defined based on an actual industrial scenario. For example, a process may be defined as corresponding to a workcell, or a workstation in the process may be defined as a workcell, or a workbench in the workstation may be defined as corresponding to a workcell, etc. Different workcells have different processes.

[0063] Service operations may be performed by different entities. The entities, for example, may be physical devices, may be persons, or may be other virtualized resources. For ease of description, the physical devices, the persons, the virtualization terminology resources, and the like may be collectively referred to as resources. The resources usually refer to various operating entities that can execute the workflow on site. In some embodiments, a resource as an operating entity may be used as a common configuration parameter of a function block node, and corresponding resource configuration may be performed on the required function block node during creation of a behavior tree. In some embodiments, during creation of the function block node, the resource as the operating entity is configured for the function block node, so that the resource configuration does not need to be performed on the required function block node during the creation of the behavior tree. In some embodiments, for ease of resource management, these resources may be represented in forms of resource nodes, and the resource nodes may be stored in forms of resource knowledge graphs. Each resource knowledge graph includes resource nodes, and a connection line representing a relationship between resource nodes.

[0064] Herein, the subject thesaurus associated with the subject placeholder may be displayed in a new pop-up window on the natural language sentence editing interface, and the subject occupying the subject placeholder may be determined from the subject thesaurus based on user selection, that is, the resource in the workcell may be determined. In some embodiments, the subject thesaurus may be displayed on the natural language sentence editing interface through a pull-down menu.

[0065] Step 3: Determine a predicate placeholder based on a second selection operation performed on the language element library. Similarly, the user may select the predicate placeholder from the language element library and drag the predicate placeholder to the natural language sentence editing interface to determine the predicate placeholder. Since dragging positions of the user are different, the predicate placeholder may be dragged to a rear of the subject placeholder or may be dragged to a front of the subject placeholder. When the predicate placeholder is dragged to the front of the subject placeholder, the predicate placeholder may be automatically adjusted to the rear of the subject placeholder, to conform to a language habit.

[0066] Step 4: Determine, from a predicate thesaurus associated with the subject, a predicate occupying the predicate placeholder, where the predicate corresponds to a service operation of the resource. Herein, the predicate thesaurus associated with the subject may be displayed in the new pop-up window on the natural language sentence editing interface, and the predicate occupying the

predicate placeholder may be determined from the predicate thesaurus based on user selection, that is, the service operation that the resource can perform. In some embodiments, the predicate thesaurus may be displayed through the pull-down menu on the natural language sentence editing interface.

[0067] Step 5: Generate the natural language sentence based on the occupied subject placeholder and the occupied predicate placeholder, where the natural language sentence is adapted to be converted into a low-code representation describing a workflow, and the workflow represents that resource performs service operation. In some embodiments, in the generated natural language sentence, a placeholder of each type does not have a visual effect, and a sentence constituent occupying the placeholder is merely displayed at a corresponding position.

[0068] The generated natural language sentence instructs the resource to perform the service operation, and therefore can be converted into the low-code representation describing the workflow. Therefore, based on the operations performed on the placeholders and the libraries, the workflow may be conveniently described as the natural language sentence by using the semantic language element, and a person without programming knowledge may conveniently describe the workflow by using the natural language sentence, which reduces implementation difficulty of the workflow. Moreover, the natural language sentence may be converted into the low-code representation that can be recognized by an underlying device, which further enriches implementations of the workflow.

[0069] In some embodiments, the method for generating a natural language sentence describing a workflow further includes at least one of the following: (1). determining an object placeholder based on a third selection operation performed on the language element library, and determining, from a BOM (bill of materials) associated with the service operation, an object occupying the object placeholder, where the object serves as an execution target of the service operation; (2). determining an adverbial placeholder based on a fourth selection operation performed on the language element library, and determining an adverbial occupying the adverbial placeholder based on input data of the service operation; (3). determining a complement placeholder based on a fifth selection operation performed on the language element library, and determining, from the BOM associated with the service operation, a complement occupying the complement placeholder, where the complement serves as a supplementary description of the service operation; (4). determining an attributive placeholder based on a sixth selection operation performed on the language element library, and determining, from a user input or the BOM associated with the service operation, an attributive occupying the attributive placeholder, where the attributive serves as a limiting description of the subject or the object; (5). determining an annotation placeholder based on a seventh selection operation performed on the language element library, and determining, from the user input, an annotation occupying the annotation placeholder; (6). determining a figure placeholder based on an eighth selection operation performed on the language element library, and determining, from the user input, a figure occupying the figure placeholder; and (7) determining a video placeholder based on a ninth selection operation performed on the language element library, and determining, from the user input, a video occupying the video placeholder.

[0070] In some embodiments, the method for generating a natural language sentence describing a workflow further includes: determining a logic placeholder and a semantic expression placeholder based on a tenth selection operation performed on the language element library; determining, from the user input, a logic keyword occupying the logic placeholder and a semantic expression occupying the semantic expression placeholder; determining a logic execution branch based on the logic keyword and the semantic expression; and determining a natural language sentence corresponding to each logic execution branch. For example, the logic keyword may be a logic keyword in condition determining (if-then-else), or a composite logic (for example, switch-case-default, fallback, or guard), or the like. A logic value (which is usually true or false) corresponding to the semantic expression is calculated, and the preset logic execution branch corresponding to the

logic value is selected by using the logic determined by the logic keyword.

[0071] In some embodiments, the method for generating a natural language sentence describing a workflow further includes: generating a label of the natural language sentence based on an execution order of the natural language sentence. The label may be used for representing an order serial number of the natural language sentence. For example, the label may be displayed at a beginning, or an end, or the like of the natural language sentence, to be used for identifying the execution order of the natural language sentence.

[0072] In some embodiments, the method for generating a natural language sentence describing a workflow further includes: determining an exception placeholder based on an eleventh selection operation performed on the language element library; determining, from the user input, a label corresponding to the exception placeholder; selecting an exception type from exception library corresponding to the exception placeholder; determining an exception execution branch when the exception type occurs during execution of a natural language sentence of the label; and determining a natural language sentence corresponding to the exception execution branch. For example, the exception type includes overheating and overvoltage. Correspondingly, the exception execution branch includes an exception processing process when the overheating occurs and an exception processing process when the overvoltage occurs.

[0073] In some embodiments, the method for generating a natural language sentence describing a workflow further includes: determining a keypoint placeholder based on a twelfth selection operation performed on the language element library; and determining, from the user input, a keypoint occupying the keypoint placeholder, where the low-code representation includes a control corresponding to the keypoint, and when the control is triggered, a next keypoint of the keypoint in the workflow is executed.

[0074] Therefore, definitions of semantic elements are further extended, to provide support for necessary low-code constituents such as a behavior tree logic, the semantic expression, the exception, the keypoint, the annotation, the figure, the video, and an index.

[0075] In some embodiments, the method for generating a natural language sentence describing a workflow further includes: receiving a change request for the occupied subject placeholder in the natural language sentence; displaying the subject thesaurus, especially automatically displaying the subject thesaurus, based on the change request; determining, by the user from the subject thesaurus, an updated subject updating the occupied subject placeholder, where the updated subject corresponds to the updated resource; displaying (for example, automatically displaying) a predicate thesaurus associated with the updated subject; determining, from the predicate thesaurus associated with the updated subject, an updated predicate updating the occupied predicate placeholder (for example, the user selects the updated predicate from the predicate thesaurus associated with the updated subject), where the updated predicate corresponds to a service operation of the updated resource; and updating (for example, automatically updating) the natural language sentence based on the updated occupied subject placeholder and the updated occupied predicate placeholder. It may be learned that, in this implementation of the present invention, the subject may be conveniently changed to change the resource in the workflow.

[0076] In some embodiments, the method for generating a natural language sentence describing a workflow further includes: receiving a change request for the occupied predicate placeholder in the natural language sentence; displaying (for example, automatically displaying) the predicate thesaurus associated with the subject based on the change request; determining, from the predicate thesaurus, an updated predicate updating the occupied predicate placeholder (for example, the user selects the updated predicate from the predicate thesaurus), where the updated predicate corresponds to an updated service operation of the resource; and updating (for example, automatically updating) the natural language sentence based on the occupied subject placeholder and the updated occupied predicate placeholder. It may be learned that, in this implementation of the present invention, the predicate may be conveniently changed to change the service operation in

the workflow.

[0077] Some embodiments include a natural language sentence conversion method. The natural language sentence determined in the above manner may be converted into an FBTD. The conversion method includes:

[0078] Step 1: Determine a first mapping relationship between a subject of a natural language sentence and a function block head of a function block, where the subject corresponds to a resource in a workcell.

[0079] Step 2: Determine a second mapping relationship between a predicate of the natural language sentence and a function block name of the function block, where the predicate corresponds to a service operation of the resource.

[0080] Step 3: Convert the natural language sentence into the function block or convert the function block into the natural language sentence based on the first mapping relationship and the second mapping relationship, where the function block is presented in an FBTD, the function block is adapted to describe a workflow, and the workflow represents the resource to perform the service operation. A process of adjusting a natural language sentence is exemplarily described by using a natural language sentence in a low-code architecture as an example.

[0081] FIG. 2A shows a schematic diagram for describing an example workflow in a low-code architecture in a no-code manner incorporating teachings of the present disclosure. In the architecture, all exchange workpieces adopt a standardized universal runtime engineering exchange format, and metadata of the universal runtime engineering exchange format is easily tracked and managed.

[0082] A universal runtime engineering exchange model **304** includes a logic library **307** (for example, the logic library may be a behavior tree logic library) of a low-code platform, a resource library **308**, a function block library **309**, a customized thesaurus **310**, a BOM **311**, a universal language extension thesaurus **312**, and a keypoint library **313**. The logic library **307** provides a logic for occupying a logic placeholder **405**, a semantic expression for occupying a semantic expression placeholder **407**, and an exception type for occupying an exception placeholder **408**. The keypoint library **313** provides a keypoint for occupying a keypoint placeholder **409**. The resource library **308** provides a resource for occupying a subject placeholder **401**. The function block library **309** provides a service operation for occupying a predicate placeholder **402** and input data of the service operation for occupying an adverbial placeholder **403**. The BOM **311** provides an object for occupying an object placeholder **404**, an attributive for occupying an attributive placeholder **412**, and a complement for occupying a complement placeholder **413**. The universal language extension thesaurus **312** provides a preposition automatically inserted into a natural language sentence.

[0083] Through a no-code universal language editor **302** in this implementation of this application that implements the method for generating a natural language sentence and the method for adjusting a natural language sentence, the natural language sentence may be generated and edited. A SOP in a natural language form may be generated by using a SOP editor without code semantics. Moreover, the generated natural language sentence (including the SOP) may be converted into a low-code behavior tree/FBTD **305**. In the low-code behavior tree, function block node is presented in a form of a label graph. In the low-code FBTD, the function block node is presented in a form of a type graph. Then the low-code represented behavior tree/the FBTD **305** is parsed by using interpreter-driven runtime **306**, to obtain a workflow represented in a form of node link assembly, and the workflow represented in the form of the node link assembly is compiled and downloaded to runtime of a main controller of a corresponding workcell, to execute the workflow. For example, the behavior tree/the FBTD **305** may be stored in a markup language, for example, an extensible markup language (XML), and may be validated by an XML Schema (XSD) prototype, to verify that an XML format of the behavior tree/the FBTD **305** is correct.

[0084] After parsing the behavior tree/the FBTD **305**, the interpreter-driven runtime **306** may

obtain the workflow represented in the form of the node link assembly, and then compile and download the workflow to the runtime of the main controller of the corresponding workcell, to execute the workflow. An implementation of the present invention further provides a natural language sentence conversion method, which may convert a natural language sentence (For example, a natural language sentence generated without a preposition added, or a natural language sentences with a preposition added) determined based on the above manner into an FBTD.

[0085] The method includes: Step (1): Determine a first mapping relationship between a subject of a natural language sentence and a function block head of a function block, where the subject corresponds to a resource. Step (2): Determine a second mapping relationship between a predicate of the natural language sentence and a function block name of the function block, where the predicate corresponds to a service operation of the resource. Step (3): Convert the natural language sentence into the function block, or convert the function block into the natural language sentence based on the first mapping relationship and the second mapping relationship, where the function block is presented in an FBTD, the function block is adapted to describe a workflow, and the workflow represents the resource to perform the service operation. The existence of the preposition is ignored during the conversion.

[0086] A specific process of converting the natural language sentence into the FBTD is exemplarily described above. A person skilled in the art may realize that the description is merely exemplary and is not intended to limit the protection scope of the implementations of the present invention. Moreover, after the natural language sentence is converted into the FBTD, the FBTD may be further converted into a behavior tree, which is not limited in this implementation of the present invention.

[0087] An OT domain generally refers to an OT, which integrates hardware and software, and detects or triggers a change or an event occurring in a process in an enterprise by directly monitoring and/or controlling a physical device (which is referred to as an OT device). The OT monitors or changes, for example, a physical state of an industrial control system (ICS) by using a computer. The ICS is a facility, a system, and a device achieved based on the computer, and is configured to remotely monitor and/or control a key industrial process, to achieve a physical function. The term “OT” is used to distinguish the ICS from a conventional IT system in terms of technical implementation and function. Currently, many IT low-code development tools or platforms exist on the market. Some of the tools are used for a usage scenario of an IoT, and target an experienced IT engineer, and it is difficult for an OT engineer and a junior IT engineer to understand paradigms of the tools. However, some of the tools are more applicable to a usage scenario of low code development in an IT domain, and cannot be well applicable to the OT domain. The above method for generating a natural language sentence generation describing a workflow in this embodiment may be used in the OT domain as a no-code OT domain development method. Correspondingly, the workflow may be an OT domain workflow. The workcell may be a workcell in the OT domain. The OT device may include but is not limited to an IoT device, a programmable logic controller (PLC), robotics, a manual process, and an industrial personal computer (IPC).

[0088] In addition, integration of the IT domain and the OT domain becomes increasingly important in a process of enterprise digitalization conversion. To integrate the IT domain and the OT domain into an ITOT system, a current problem that needs to be resolved urgently is how an enterprise collects OT domain data and controls an OT domain process in a manner that is easily understood and is not IT domain programming. Therefore, the above method for generating a natural language sentence generation describing a workflow in this embodiment may be used in the ITOT system as a no-code OT domain development method that can be integrated with the IT domain.

[0089] FIG. 2B is a schematic diagram of an interaction based on a language element library incorporating teachings of the present disclosure. In FIG. 2B, a language element library **400** is

displayed. The language element library **400** includes a subject placeholder **401**, a predicate placeholder **402**, an adverbial placeholder **403**, an object placeholder **404**, a logic placeholder **405**, an annotation placeholder **406**, a semantic expression placeholder **407**, an exception placeholder **408**, a keypoint placeholder **409**, a figure placeholder **410**, a video placeholder **411**, an attributive placeholder **412**, and a complement placeholder **413**. The interaction process includes automatic label generation processing **415** and automatic preposition generation processing **416**. The automatic preposition generation processing **416** may generate a preposition for a natural language sentence to improve readability. The automatic label generation processing **415** automatically generates a label **417** for the natural language sentence based on an execution order of the natural language sentence. In this embodiment, the label **417** is a serial number label.

[0090] A user drags any placeholder from the language element library **400** to a natural language sentence editing interface. As shown in FIG. 2B, the user first drags the subject placeholder **401**, and the label **417** for the natural language sentence is automatically generated. In this case, a subject thesaurus **418** may be automatically displayed or displayed through triggering of the user, to help the user to select a subject occupying the subject placeholder **401** from the subject thesaurus **418**. Assuming that the user selects a “robot 3”, the “robot 3” occupies the subject placeholder **401**, and a natural language sentence **419** after the subject is determined is obtained. Then the user drags the predicate placeholder **402** from the language element library **400** after the “robot 3” of the natural language sentence **419** after the subject is determined and obtains a natural language sentence **420** after the predicate placeholder is dragged. A predicate thesaurus **421** associated with the “robot 3” (that is, a service operation that the “robot 3” may perform) is displayed, for the user to select a predicate occupying the predicate placeholder **402** from the predicate thesaurus **421**. Assuming that the user selects “linearly move”, the “linearly move” occupies the predicate placeholder **402**, and a natural language sentence **422** “A robot 3 linearly moves” after the predicate is determined is obtained.

[0091] In this case, the user may click/tap the “robot 3” to re-display the subject thesaurus **418**, to help the user change the subject. Correspondingly, the user may click/tap the “linearly move” to re-display the predicate thesaurus **421**, to help the user change the predicate.

[0092] A universal language word order template for a SOP may be established based on this implementation of the present invention. FIG. 2C is a schematic diagram of a universal language word order template of a SOP according to an embodiment of this application. A template **431** includes a plurality of natural language sentences determined based on the interaction process shown in FIG. 2B, and each of the natural language sentences has a label to distinguish between execution orders. Natural language sentences including all of the execution orders may be combined into a complete SOP. In FIG. 2C, a label E1 represents exception processing. The exception processing is configured to automatically perform processing on an exception during execution of a natural language sentence corresponding to a label **2**. The exception processing process includes a natural language sentence with a label of E1.1 and a natural language sentence with a label of E1.2.

[0093] FIG. 2D is a schematic diagram of an architecture for automatically generating a preposition incorporating teachings of the present disclosure. In FIG. 2D, a process of automatically generating the preposition is described by using a SOP template **431** as an example. A language specific grammar **436** determines to use a Chinese semantic thesaurus **437** including Chinese prepositions (for example, custom-character, custom-character, custom-character, custom-character, custom-character, custom-character, custom-character, custom-character, custom-character, custom-character) or an English semantic thesaurus **438** including English prepositions (for example, from, in, at, for, on, into, by, when). A rule-based linking engine **432** parses the SOP template **431**. The linking engine **432** automatically analyzes placeholders at different positions in the SOP template **431** to generate prepositions **433** included in a determined thesaurus (the Chinese semantic thesaurus **437** or the English semantic thesaurus **438**). The

preposition **433** includes a prepositional preposition **434** and a postpositional preposition **435**. When the linking engine **432** analyzes a complement, a preposition-complement structure may be adopted, to generate the preposition before the complement, for example, “to a detection station”. When the linking engine **432** analyzes an adverbial, a preposition-adverbial structure may be adopted, to generate the preposition before the adverbial, for example, “with a torque of 450 N”. When the linking engine **432** analyzes an object, a preposition-object structure may be adopted, to generate the preposition before the object, for example, “to a circuit board”. When the linking engine **432** analyzes a prepositional object (for example, a user drags an object placeholder to a beginning of a sentence), an object-preposition structure may be adopted, to generate the preposition after the object, for example, “a circuit board is”.

[0094] FIG. 2E is a schematic diagram of automatically generating a preposition incorporating teachings of the present disclosure. In FIG. 2E, for a natural language sentence with a first label of 1.1, “Employee use torque wrench **450** torques twist screw circuit board” may be obtained through an occupation operation performed on a placeholder. Through analysis of types and positions of placeholders, a preposition “use” may be automatically generated between an “employee” and a “torque wrench”, a preposition “at” may be automatically generated between the “torque wrench” and “450 torques”, and a preposition “to” can be automatically generated between a “screw” and a “circuit board”. A user may move a placeholder based on a personal habit to change a position of the placeholder. Correspondingly, the preposition is updated based on the updated position of the placeholder. In FIG. 2E, four natural language sentences with the label of 1.1 have the same meaning and may be converted into the same low-code representation. In some embodiments, complement-to-subject conversion processing **450** may be performed. In the complement-to-subject conversion processing **450**, when an automation upgrade is completed, the complement may be converted into the subject. For example, when a robot replaces human, an end executor (for example, a torque wrench) may perform an action without human assistance. Therefore, the torque wrench serves as a body of a controller, and therefore is referred to as a subject. In this case, the original subject (“employee 1”) and the preposition (that is, “by using”) before the complement converted into the subject are deleted.

[0095] FIG. 2F is a schematic diagram of an example overall semantic editing process incorporating teachings of the present disclosure. The overall semantic editing process includes:

[0096] First step: Perform subject placeholder dragging processing **750**. A subject placeholder **401** is dragged from a language element library to a natural language sentence editing interface.

Moreover, a label **415** is generated at a fixed position (for example, a beginning of a sentence).

[0097] Second step: Perform subject determining and predicate placeholder dragging processing **751**. A subject thesaurus including each resource is displayed, for a user to select and drag a subject (a resource) occupying the subject placeholder **401** from the subject thesaurus. For example, a “robot 1” as the subject is dragged from the subject thesaurus including a large number of resources. Then a predicate placeholder **402** (which is usually dragged after the subject) is dragged from the language element library.

[0098] Third step: Perform predicate determining processing **752**. A predicate thesaurus associated with the “robot 1” is displayed, the predicate thesaurus associated with the “robot 1” includes all service operations that the “robot 1” can perform. The user selects and drags a service operation occupying the predicate placeholder **402** from the predicate thesaurus associated with the “robot 1”. For example, the dragged predicate is picking and placing.

[0099] Fourth step: Perform object placeholder dragging and first preposition generation processing **753**. An object placeholder **404** is dragged from the language element library to the natural language sentence editing interface. Based on the dragging operation performed by the user, a position of the object placeholder **404** may not be fixed. Moreover, depending on the position of the object placeholder **404**, a preposition related to the object is automatically generated. In this case, overall semantics corresponding to each preposition remain the same. For example, when the

object placeholder **404** is located after the subject and the predicate, no preposition is generated. When the object placeholder **404** is located between the subject and the predicate, a preposition “by” is generated between the object placeholder **404** and the subject. When the object placeholder **404** is located before the subject, a preposition “is/are” is generated between the object placeholder **404** and the subject.

[0100] Fifth step: Perform object determining processing **754**. A to-be-picked object may be determined as a circuit board based on a BOM, and the circuit board may be determined as an object filling the object placeholder **404**.

[0101] Sixth step: Perform adverbial placeholder dragging, adverbial determining, and second preposition generation processing **755**. One or more adverbial placeholders are dragged from the language element library to the natural language sentence editing interface. Moreover, an adverbial filling the adverbial placeholder may be determined through input data of the service operation or a user input, and a preposition related to the adverbial is generated. For example, it is assumed that the input data of the service operation “pick and place” includes two parameters, namely, a speed and an acceleration. Specific values of the speed and the acceleration may be provided based on the user input (assuming that the speed is 50 m/s and the acceleration is 120 m/s.^{sup.2}), the adverbials include the “speed of 50 m/s” and the “acceleration of 120 m/s.^{sup.2}”, and prepositions “at” are respectively generated before the adverbials.

[0102] Seventh step: Perform complement placeholder dragging, complement determining, and second preposition generation processing **756**. One or more complement placeholders are dragged from the language element library to the natural language sentence editing interface (for example, an end of the sentence). Moreover, a complement filling the complement placeholder may be determined through output data of the service operation or the user input, and a preposition related to the complement is generated. For example, assuming that the user input includes a “detection station” as a starting point object of the service operation (for example, “pick and place”) and a “tray A” as an ending point object, the complements include the “detection station” and the “tray A”, and a preposition “from” corresponding to the starting point object is generated before the “detection station”, and a preposition “to” corresponding to the ending point object is generated before the “tray A”.

[0103] Eighth step: Perform attributive placeholder dragging and attributive determining processing **757**. One or more attributive placeholders (which are usually dragged before a noun to limit the noun) are dragged from the language element library to the natural language sentence editing interface. Moreover, an attributive filling the attributive placeholder may be determined through a BOM or the user input.

[0104] Ninth step: Perform word order adjustment processing **758**. In this case, each placeholder may be dragged, and the automatically generated preposition is adjusted based on the updated placeholder position. For example, when the object is prepositioned to form a passive voice, a sentence is more fluent through addition of the preposition “be”.

[0105] FIG. **3** is an application scenario of a method for adjusting a natural language sentence in an industrial automation field incorporating teachings of the present disclosure. In FIG. **3**, a low-code development tool **10** generates a natural language sentence describing an OT domain workflow after a user performs a selection operation, a dragging operation, and a filling operation on placeholders in the language element library. Moreover, the low-code development tool **10** automatically adds a preposition in the natural language sentence based on a type attribute of a placeholder and a position of the placeholder in the natural language sentence, to adjust the natural language sentence, for ease of user understanding. The OT domain workflow is defined as operations to be performed by a production line as a workcell, as shown on a right side of FIG. **3**. The low-code development tool **10** converts the natural language sentence into a behavior tree corresponding to the OT domain workflow. The behavior tree is posted to runtime **30**, so that the runtime **30** controls production line operations completing the workcell. In addition, a

corresponding microservice may be generated by the microservice generator **20** based on the behavior tree and registered in a knowledge platform **200**. In this way, a code development tool **301** of an IT domain may call the corresponding microservice through the knowledge center **200**. [0106] A user may select a placeholder, drag the placeholder, or enter content in the placeholders at a lower left corner GUI as shown in FIG. 3, to edit the natural language sentence. For example, required data (for example, a workpiece processing parameter) is obtained first from a database and a server through the knowledge platform **200**, and operation of an entire workcell is controlled. The workcell herein is a production line. The production line includes a machine, a conveyor belt, a robot arm, a person, a programmable logic controller (PLC), an accessory gear box (AGB), and the like. In a specific implementation, the code development tool **301** in the IT domain may be located on the same hardware device as the low-code development tool **10**, for example, on a same computer.

[0107] The application scenario of adjusting a natural language sentence is described above by using the industrial automation field as an example. A person skilled in the art may realize that the description herein is merely exemplary and is not intended to limit the protection scope of the implementations of the present disclosure.

[0108] FIG. 4 is a structural diagram of an example apparatus for adjusting a natural language sentence incorporating teachings of the present disclosure. The apparatus **700** for adjusting a natural language sentence includes: a receiving module **701**, configured to receive a natural language sentence including a placeholder, where the placeholder includes a type attribute, and the placeholder is occupied by a sentence constituent matching the type attribute; a determining module **702**, configured to determine a position of the placeholder in the natural language sentence; and an addition module **703**, configured to add a preposition to the natural language sentence based on the type attribute of the placeholder and the position of the placeholder in the natural language sentence.

[0109] In some embodiments, the addition module **703** is configured to: determine a first sentence constituent occupying subject placeholder in the natural language sentence as a subject of the natural language sentence; determine a second sentence constituent occupying a predicate placeholder in the natural language sentence as a predicate of the natural language sentence; determine a third sentence constituent occupying an object placeholder in the natural language sentence as an object in the natural language sentence; add a first preposition between the object and the subject when the object is located between the subject and the predicate; and add a second preposition between the object and the subject when the object is located before the subject and the predicate.

[0110] In some embodiments, the addition module **703** is configured to: determine a fourth sentence constituent occupying a complement placeholder in the natural language sentence as a first complement of the natural language sentence, where the first complement is determined based on a user input; and add a third preposition before the first complement.

[0111] In some embodiments, the addition module **703** is configured to: determine a fifth sentence constituent occupying the complement placeholder in the natural language sentence as a second complement of the natural language sentence, where the second complement is determined based on output data of a service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and add a fourth preposition before the second complement.

[0112] In some embodiments, the addition module **703** is configured to: determine a sixth sentence constituent occupying an adverbial placeholder in the natural language sentence as an adverbial of the natural language sentence, where the adverbial is determined based on input data of the service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and add a fifth preposition before the adverbial.

[0113] In some embodiments, the apparatus further includes an update module **704**, configured to:

move the placeholder in natural language sentence based on a user-triggered instruction; determine an updated position of the placeholder in the natural language sentence; and update the preposition in the natural language sentence based on the type attribute of the placeholder and the update position of the placeholder in the natural language sentence.

[0114] In some embodiments, the natural language sentence is adapted to describe a workflow. The workflow represents that a resource in a workcell performs the service operation. In some embodiments, the apparatus further includes an execution module **705**, configured to: convert natural language sentence into a low-code representation describing workflow; parse the low-code representation, to obtain a workflow represented in a form of node link assembly; and compile and download the workflow represented in the form of the node link assembly to runtime of a main controller of a corresponding workcell, to execute the workflow.

[0115] FIG. **5** is a structural diagram of an example apparatus with a memory-processor architecture for adjusting a natural language sentence incorporating teachings of the present disclosure. The apparatus **500** for adjusting a natural language sentence includes at least one memory **501** and at least one processor **502**. The at least one processor **502** is configured to invoke a computer program stored in the at least one memory **501**, to perform one or more of the methods for adjusting a natural language sentence described herein.

[0116] In some embodiments, a storage medium stores computer-readable code for implementing the functions of any of the above implementations, and enables a computer (or a central processing unit (CPU) or a microprocessor unit (MPU)) of the system or the device to read and execute the computer-readable code stored in the storage medium. In addition, an operating system or the like running on the computer may be enabled through computer-readable code-based instructions to complete some or all actual operations. The computer-readable code read from the storage medium may also be written into a memory arranged in an extension board inserted in the computer, or may be written into a memory arranged in an extension unit connected to the computer, and then a CPU or the like mounted to the extension board or the extension unit may be enabled to execute some or all of the actual operations through the computer-readable code-based instructions, to implement the functions of any of the above implementations.

[0117] Examples of the computer-readable medium include, but are not limited to, a floppy disk, a portable compact disk read-only memory (CD-ROM), a magnetic disk, an optical disk (for example, the CD-ROM, a compact disk-recordable (CD-R), a compact disk-rewritable (CD-RW), a digital video disk-read only memory (DVD-ROM), a digital video disk-random access memory (DVD-RAM), a digital video disk-rewritable (DVD-RW), a DVD+RW), a memory chip, a ROM, a RAM, an application specific integrated circuit (ASIC), a configured processor, an all-optical medium, all magnetic tapes, or another magnetic medium, or any another medium from which a computer processor may read instructions. In addition, computer-readable media in various other forms may send or carry instructions to the computer, including a router, a private or public network, or another wired and wireless transmission device or channel. For example, computer-readable instructions may be downloaded from a server computer or a cloud through a communication network. The instructions may include code of any computer programming language, including C, C++, C language, Visual Basic, java, and JavaScript.

[0118] It should be noted that, not all steps and modules in the above processes and diagrams of the system structures are necessary, and some steps or modules may be omitted based on an actual requirement. An execution order of the steps is not fixed and may be adjusted as required. The system structure described in the above embodiments may be a physical structure or a logical structure. To be specific, some modules may be implemented by the same physical entity, or some modules may be implemented by a plurality of physical entities or may be jointly implemented by some components in a plurality of independent devices.

[0119] The above descriptions are merely example embodiments of the teachings of the present disclosure and are not intended to limit the scope thereof. Any modification, equivalent

replacement, or improvement made within the spirit and principle of the present disclosure shall fall within the protection scope.

Claims

1. A method for adjusting a natural language sentence, the method comprising: receiving or obtaining a natural language sentence comprising a placeholder, wherein the placeholder comprises a type attribute, and the placeholder is occupied by a sentence constituent matching the type attribute; determining a position of the placeholder in the natural language sentence; and adding a preposition to the natural language sentence based on the type attribute of the placeholder and the position of the placeholder in the natural language sentence.
2. The method according to claim 1, wherein adding a preposition to the natural language sentence based on a type of the placeholder and the position of the placeholder in the natural language sentence comprises: determining a first sentence constituent occupying a subject placeholder in the natural language sentence as a subject of the natural language sentence; determining a second sentence constituent occupying a predicate placeholder in the natural language sentence as a predicate of the natural language sentence; determining a third sentence constituent occupying an object placeholder in the natural language sentence as an object of the natural language sentence; adding a first preposition between the subject and the object when the object is located between the subject and the predicate; and adding a second preposition between the object and the subject when the object is located before the subject and the predicate.
3. The method according to claim 1, wherein adding a preposition to the natural language sentence based on a type of the placeholder and the position of the placeholder in the natural language sentence comprises: determining a fourth sentence constituent occupying a complement placeholder in the natural language sentence as a first complement of the natural language sentence, wherein the first complement is determined based on a user input; and adding a third preposition before the first complement.
4. The method according to claim 1, wherein adding a preposition to the natural language sentence based on a type of the placeholder and the position of the placeholder in the natural language sentence comprises: determining a fifth sentence constituent occupying the complement placeholder in the natural language sentence as a second complement of the natural language sentence, wherein the second complement is determined based on output data of a service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and adding a fourth preposition before the second complement.
5. The method according to claim 1, wherein adding a preposition to the natural language sentence based on a type of the placeholder and the position of the placeholder in the natural language sentence comprises: determining a sixth sentence constituent occupying an adverbial placeholder in the natural language sentence as an adverbial of the natural language sentence, wherein the adverbial is determined based on input data of the service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and adding a fifth preposition before the adverbial.
6. The method according to claim 1, further comprising: moving the placeholder in the natural language sentence based on a user-triggered instruction; determining an updated position of the placeholder in the natural language sentence; and updating the preposition in the natural language sentence based on the type attribute of the placeholder and the updated position of the placeholder in the natural language sentence.
7. The method according to claim 1, wherein: the natural language sentence describes a workflow; and the workflow represents a resource in a workcell performing the service operation.
8. The method according to claim 7, further comprising: converting the natural language sentence

into a low-code representation describing the workflow; parsing the low-code representation, to obtain a workflow represented in a form of node link assembly; and compiling and downloading the workflow represented in the form of the node link assembly to runtime of a main controller of a corresponding workcell, to execute the workflow.

9. An apparatus for adjusting a natural language sentence, the apparatus comprising: a receiving module to receive a natural language sentence comprising a placeholder, wherein the placeholder comprises a type attribute, and the placeholder is occupied by a sentence constituent matching the type attribute; a determining module to determine a position of the placeholder in the natural language sentence; and an addition module to add a preposition to the natural language sentence based on the type attribute of the placeholder and the position of the placeholder in the natural language sentence.

10. The apparatus according to claim 9, wherein the addition module is configured to: determine a first sentence constituent occupying a subject placeholder in the natural language sentence as a subject of the natural language sentence; determine a second sentence constituent occupying a predicate placeholder in the natural language sentence as a predicate of the natural language sentence; determine a third sentence constituent occupying an object placeholder in the natural language sentence as an object of the natural language sentence; add a first preposition between the subject and the object when the object is located between the subject and the predicate; and add a second preposition between the object and the subject when the object is located before the subject and the predicate.

11. The apparatus according to claim 9, wherein the addition module is configured to: determine a fourth sentence constituent occupying a complement placeholder in the natural language sentence as a first complement of the natural language sentence, wherein the first complement is determined based on a user input; and add a third preposition before the first complement.

12. The apparatus according to claim 9, wherein the addition module is configured to: determine a fifth sentence constituent occupying the complement placeholder in the natural language sentence as a second complement of the natural language sentence, wherein the second complement is determined based on output data of a service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and add a fourth preposition before the second complement.

13. The apparatus according to claim 9, wherein the addition module is configured to: determine a sixth sentence constituent occupying an adverbial placeholder in the natural language sentence as an adverbial of the natural language sentence, wherein the adverbial is determined based on input data of the service operation, and the service operation is determined based on the predicate occupying the predicate placeholder in the natural language sentence; and add a fifth preposition before the adverbial.

14. The apparatus according to claim 9, the apparatus further comprising an update module configured to: move the placeholder in the natural language sentence based on a user-triggered instruction; determine an updated position of the placeholder in the natural language sentence; and update the preposition in the natural language sentence based on the type attribute of the placeholder and the updated position of the placeholder in the natural language sentence.

15. The apparatus according to claim 9, wherein the natural language sentence is adapted to describe a workflow, and the workflow represents that a resource in workcell performs the service operation.

16. The apparatus according to claim 15, further comprising an execution module configured to: convert the natural language sentence into a low-code representation describing the workflow; parse the low-code representation, to obtain a workflow represented in a form of node link assembly; and compile and download the workflow represented in the form of the node link assembly to runtime of a main controller of a corresponding workcell, to execute the workflow.

17-19. (canceled)

